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23 August 1984
SUPERSEDING
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12 November 1980

MILITARY SPECIFICATION

ERECTION AND MAINTENANCE EQUIPMENT FOR CABLE REINFORCEMENT SET FOR PANEL BRIDGE, M-2 (BAILEY TYPE)

This specification is approved for use by the USA Belvoir Research and Development Center, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers all equipment required to install and maintain the cable reinforcement equipment for panel bridge M-2 (Bailey type) conforming to MIL-E-52714(ME).

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. Unless otherwise specified (see 6.2), the following specifications and standards of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DoDISS) specified in the solicitation, form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

QQ-S-781	- Strapping, Steel and Seals.
PPP-B-601	- Boxes, Wood, Cleated-Plywood.
PPP-B-621	- Boxes, Wood, Nailed and Lock-Corner.
PPP-T-60	- Tape, Packaging, Waterproof.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: USA Belvoir Research and Development Center, ATTN: STRBE-DS, Fort Belvoir, VA 22060-5606 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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MIL-P-116	- Preservation, Methods of.
MIL-B-121	- Barrier Material, Greaseproofed, Waterproofed, Flexible.
MIL-P-514	- Plate, Identification, Instruction and Marking, Blank.
MIL-T-704	- Treatment and Painting of Materiel.
MIL-C-12044	- Chest, Plywood.
MIL-C-16173	- Corrosion Preventive Compound, Solvent Cutback, Cold Application.
MIL-G-20241	- Gasket Material, Wool Felt, Impregnated, Adhesive, Pressure-Sensitive.
MIL-B-43436	- Band, Marker, Blank.
MIL-L-46167	- Lubricating Oil, Internal Combustion Engine, Arctic.
MIL-E-52714	- Equipment, Cable Reinforcement, for Panel Bridge, M-2 (Bailey Type).
MIL-C-52950	- Crate, Wood, Open and Covered.
MIL-S-81733	- Sealing and Coating Compound, Corrosion Inhibitive.

STANDARDS

MILITARY

MIL-STD-105	- Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129	- Marking for Shipment and Storage.
MIL-STD-889	- Dissimilar Metals.
MIL-STD-1186	- Cushioning, Anchoring, Bracing, Blocking, and Waterproofing with Appropriate Test Methods.
MIL-STD-1261	- Welding Procedures for Constructional Steels.

2.1.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this specification to the extent specified herein.

DRAWINGS

ME

TA13217E2171	- Erection Equipment for Set, Cable Reinforcement, for Panel Bridge M-2 (Bailey Type).
TA13217E2172	- Maintenance Equipment for Set, Cable Reinforcement, for Panel Bridge M-2 (Bailey Type).

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PUBLICATIONS

TECHNICAL MANUALS

TM 5-277 - Bailey Bridge.

(Copies of specifications, standards, and drawings required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

2.2 Other publications. The following document(s) form a part of this specification to the extent specified herein. The issues of the documents which are indicated as DoD adopted shall be the issue listed in the current DoDISS and the supplement thereto, if applicable.

AMERICAN NATIONAL STANDARDS INSTITUTE (ANSI)

ANSI/ASME - Boiler and Pressure Vessel Code, Section IX, Welding Qualifications.
ANSI/AWS A5.1 - Electrodes, Arc Welding, Covered Carbon Steel.
ANSI/AWS A5.5 - Electrodes, Arc Welding, Low Alloy Steel Covered.
ANSI/AWS A5.17 - Electrode and Fluxes, Carbon Steel, for Submerged Arc Welding.
ANSI/AWS A5.18 - Welding, Carbon Steel Filler Metals for Gas Shielded Arc.
ANSI/AWS D1.1 - Structural Welding Code - Steel, Section 5, Qualification.

(Application for copies should be addressed to the American National Standards Institute, Inc., 1430 Broadway, New York, NY 10018.)

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

ASTM D 3951 - Standard Practice for Commercial Packaging.

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

(Industry association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

3. REQUIREMENTS

3.1 Description. The erection equipment shall be in accordance with TA13217E2171 and as specified herein. Maintenance equipment shall be in accordance with TA13217E2172 and as specified herein. The contractor shall furnish all the erection equipment, all the maintenance equipment, or separate individual components selected from either of the two top assemblies as specified (see 6.2).

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3.1.1 Drawings. The drawings forming a part of this specification are end product drawings. No deviation from the prescribed dimensions or tolerances is permissible without prior approval of the contracting officer. Where tolerances could cumulatively result in incorrect fits, the contractor shall provide tolerances within those prescribed on the drawings to insure correct fit, assembly, and operation of the end product. Any data (e.g. shop drawings, layouts, flow sheets, processing procedures, etc.) prepared by the contractor or obtained from a vendor to support fabrication and manufacture of the production item shall be made available, upon request for inspection by the contracting officer or the designated representative.

3.2 First article. When specified (see 6.2), a sample shall be subjected to first article inspection (see 4.3 and 6.3). Any changes or deviations of production equipment and components from the approved first article during production will be subject to the approval of the contracting officer. Approval of the first article will not relieve the contractor of his obligation to furnish equipment and components conforming to this specification.

3.2.1 Hydraulic-system components. In addition to the first article, all hydraulic components shown on the drawings shall be furnished and tested by the contractor prior to the time the first article is furnished. Tests shall be those specified in 4.5.2.1, 4.5.2.2, 4.5.2.3 and 4.5.2.5 as applicable and shall be subject to surveillance and approval of the Government.

3.3 Material. Material shall be as specified herein and on the applicable drawings. Materials not specified shall be selected by the contractor and shall be subject to all provisions of this specification.

3.3.1 Material deterioration and control. The erection and maintenance equipment shall be fabricated from compatible materials, inherently corrosion and deterioration resistant or treated to provide protection against the various forms of corrosion and deterioration that may be encountered in any of the applicable storage and operating environment to which the item may be exposed.

3.3.1.1 Dissimilar metals. Dissimilar metals, as defined in MIL-STD-889, shall be electrically insulated from one another to minimize or prevent galvanic corrosion. Insulation may be provided by an insulating barrier such as a corrosion inhibiting sealant conforming to MIL-S-81733 or chromate tape conforming to MIL-G-20241. Protection against any galvanic corrosion could also be obtained by exclusion of the electrolyte if feasible.

3.3.1.2 Identification of materials and finishes. The contractor shall identify the specific material, material finish or treatment for use with components and sub-components, and shall make information available, upon request, to the contracting officer or designated representative.

3.3.2 Recovered materials. For the purpose of this requirement, recovered materials are those materials which have been collected from solid waste and reprocessed to become a source of raw materials, as distinguished from virgin raw materials. The components, pieces and parts incorporated in the erection and maintenance equipment may be newly fabricated from recovered materials to

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the maximum extent practicable, provided the erection and maintenance equipment produced meets all other requirements of this specification. Used, rebuilt or remanufactured components, pieces and parts shall not be incorporated in the erection and maintenance equipment.

3.3.3 Hydraulic oil. Hydraulic oil shall conform to MIL-L-46167.

3.3.4 Polytetraflorethylene tape. Polytetraflorethylene tape shall be commercial type and shall withstand temperatures in the range from -65° F to +200° F without deterioration. The tape shall be the type that allows threaded connections to be disconnected for maintenance.

3.3.5 Welding electrodes.

3.3.5.1 High strength steel. Welding electrodes for high strength steel shall be as specified on the drawings.

3.3.5.2 Low-alloy and structural steel. In addition to the welding electrodes specified on the drawings, welding electrodes for low-alloy and structural steel shall conform to either of the following AWS specifications:

Specification	Classification
A5.1, A5.5, A5.17, A5.18	E60XX, as applicable

3.4 Hydraulic system.

3.4.1 Component cleanliness. All hydraulic components, including hose assemblies, shall be cleaned prior to assembly by flushing.

3.4.2 Hydraulic cylinders. When tested in accordance with 4.5.2.3 and 4.5.2.4 the hydraulic cylinders shall show no evidence of external leakage (except weepage normally expected at piston rod packing), permanent deformation, or rupture of any part.

3.4.3 Hydraulic hose and hose and coupling assemblies.

3.4.3.1 Hydraulic hose. Hydraulic hose shall be not more than four calendar quarters old at the time the hose and coupling assemblies are delivered to the Government.

3.4.3.2 Hose and coupling assemblies. When tested in accordance with 4.5.2.3 the hose assemblies shall withstand a 15,000 psi proof pressure without permanent deformation, rupture of any part, or external leakage.

3.4.3.3 Hose and coupling assemblies, male and female. Hose and coupling assemblies, male and female, shall withstand 15,000 psi proof pressure without permanent deformation, rupture of any part, or external leakage when tested in accordance with 4.5.2.3.

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3.4.4 Cable tensioning assembly. The cable tensioning assembly shall be assembled in accordance with Drawing 13217E2193 and when tested in accordance with 4.5.2.4 shall show no evidence of external leakage (except weepage normally expected at piston rod packing), permanent deformation, or rupture of any part.

3.4.4.1 High and low temperature. The cable tensioning assembly shall withstand high and low temperature testing as specified in 4.5.2.7 without evidence of equipment malfunction, external leakage of a hydraulic component or permanent deformation or damage.

3.4.5 Hydraulic couplings and fittings. Hydraulic couplings and fittings including bushings, flow controls, pipe nipples, elbows, and gage adapters shall show no evidence of external leakage, permanent deformation, or rupture of any part when tested in accordance with 4.5.2.3.

3.4.6 Hydraulic hand pump. The hydraulic hand pump shall be assembled in accordance with Drawing 13217E2198 and shall show no evidence of external leakage, permanent deformation, or rupture of any part when tested in accordance with 4.5.2.5.

3.5 Sealing of threaded hydraulic connections. Each connection of hydraulic fittings with pipe threads shall be sealed with polytetraflorethylene tape.

3.6 Carrying chests. When all erection equipment shown on TA13217E2171 or all maintenance equipment shown on TA13217E2172 is specified (see 3.1), the following chests shall be furnished:

Erection Equipment, Chest 1 (Two Required)

Drawing	Quantity	Description
13225E8945	1 each	Assembly, Power Unit
13220E6842	1 each	Hose and Coupling Assembly, Female
13217E2205-2	1 each	Hose and Coupling Assembly, Male

Erection Equipment, Chest 2 (Two Required)

Drawing	Quantity	Description
13217E2195	3 each	Cylinder Assembly, Hydraulic, Double Acting
13217E2196-1	2 each	Hose and Coupling Assembly, Female
13217E2205-1	2 each	Hose and Coupling Assembly, Male

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Maintenance Equipment, Chest 3 (One Required)

Drawing	Quantity	Description
13217E2195	1 each	Cylinder Assembly, Hydraulic, Double Acting
13217E2190	1 each	Adapter
13220E6842	1 each	Hose and Coupling Assembly, Female
13217E2205-2	1 each	Hose and Coupling Assembly, Male
13217E2198	1 each	Pump Assembly, Hydraulic, Manual
13218E9858	1 each	Nut, Split
13218E9859	1 each	Sleeve

Unless otherwise specified, carrying chests for the following separately procured components shall be furnished:

a. Assembly Power unit (Drawing 13225E8945), chest No. 1 as specified above.
b. Cylinder assembly, hydraulic, double acting (Drawing 13217E2195), chest No. 2 as specified above, including sufficient space for hose and coupling assemblies.

All chests shall conform to MIL-C-12044. Each chest shall separately accommodate the components in as compact a manner possible. The contractor shall provide adequate blocking and bracing within the chest to insure that the chests will withstand the test specified in 4.6.1.2 without damage to or shifting of any item within the chest. When specified (see 6.2), chests shall not be furnished.

3.7 Instruction plates. Each hydraulic cylinder, hydraulic power unit, and manual hydraulic pump shall be equipped with instruction plates or diagrams, including warnings and cautions, describing any special or important procedures to be followed in assembling, operating, or servicing the equipment or components. Instruction plates shall conform to MIL-P-514, type III, composition C of type I, grade A, class 1 material .020- or .032-inch thick. Plates shall be attached by screws or rivets in conspicuous places.

3.8 Stenciling. Each component, except pins, bolts, and washers, shall be stenciled with the following information:

National Stock Number
Nomenclature
Contract Number

Stenciling shall be in accordance with MIL-T-704 and shall be Gothic capitals and numerals, 2 inches high where space permits. Stenciling on small components shall be as large as practicable.

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3.9 Metal stamping. In addition to stenciling, the national stock number, nomenclature, contract number shall be metal stamped in an exposed location near one end of each component. The letters and numerals shall be 1 inch high where space permits. On small components the metal stamping shall be as large as practicable.

3.10 Identification marking for hose assemblies. Each hose assembly shall have identification marking metal die stamped on an aluminum band conforming to MIL-B-43436, except that the band shall be of sufficient width to allow a minimum of three lines of 1/8-inch metal stamping. The marking band shall contain the following information:

National Stock Number
Nomenclature
Mfg Code _____ Date of Mfg _____

3.11 Erection equipment. The cable reinforcement erection equipment shall conform to top assembly TA13217E2171. When tested as specified in 4.5.2.6 and 4.5.2.7, the erection equipment shall show no evidence of external leakage of a hydraulic component or evidence of permanent deformation or damage.

3.12 Maintenance equipment. The cable reinforcement maintenance equipment shall conform to TA13217E2172. When tested as specified in 4.5.2.6 and 4.5.2.7, the maintenance equipment shall show no evidence of external leakage of a hydraulic component and shall show no evidence of permanent deformation or damage.

3.13 Treatment and painting. The erection and maintenance equipment shall be treated and painted in accordance with MIL-T-704, type F. Pins, bolts, nuts, and threaded surfaces shall not be painted but shall be coated with a preservative conforming to MIL-C-16173, grade 2.

3.14 Government-furnished property. When specified (see 6.2), the following property in the quantities indicated will be furnished by the Government (see 6.2):

Item No.	Nomenclature	National Stock Number	Quantity for each
1.	Bridge, Fixed, Steel Panel, Bailey Type M2.	5420-00-530-3784	1
2.	Bridge Erection Set, Fixed Bridge, Bailey Type, M2.	5420-00-530-3785	1
3.	Truck, Dump, 5 Ton, 6 x 6, M51 (or equivalent).	2320-00-835-8336	30
4.	Trailer, basic utility, pole type, 2-1/2 T, 2 whl (or equivalent).	2330-00-697-8102	9

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3.15 Workmanship.

3.15.1 Metal fabrication. The surfaces of all complete members used in fabrication shall be free from buckling, kinks, wrinkles, washboarding, burrs, scribe lines, and scratches. Corners shall be square and true. All bends shall be made with controlled means to insure uniformity of size and shape. Moving parts shall operate without binding. The straightening of material shall be done by methods that will not cause injury to the material.

3.15.2 Welders. Before assigning any welder to manual welding work covered by this specification, the contractor shall obtain certification that the welder has passed qualification tests as prescribed by either of the following listed codes for the type of welding operations to be performed and that such qualification is effective as defined by the particular code:

ANSI/ASME Boiler and Pressure Vessel Code, Section IX, Welding Qualifications.
ANSI/AWS D1.1, Structural Welding Code-Steel, Section 5, Qualification.

Contractors who make only horizontal welds need not qualify welders for "all position welding". Subject to approval by the government, the contractor's standard welder qualification may be substituted in lieu of the above codes provided that the contractor's procedure is equivalent to the above codes. The contractor shall be responsible for determining that automatic welding equipment operators are capable of producing quality welds in accordance with ANSI/ASME or ANSI/AWS codes. The certification shall be made available for review upon request of the contracting officer or the contracting officer's representative.

3.15.3 Welding. The surfaces of parts to be welded shall be free from rust, scale, paint, grease, mill scale and other foreign matter that can be removed by chipping and wire brushing. Parent materials, weld filler metals, and fabrication techniques shall be as required to enable the erection and maintenance equipment to conform to the examination and test requirements specified in section 4. Parts to be joined by fillet welds shall be brought into as close contact as possible and in no event shall be separated by more than 1/16 of an inch, unless appropriate bridging techniques are used.

3.15.3.1 High strength low-alloy steel and heat treated high strength steel. Welding of high strength steel and heat treated high strength steel shall be in accordance with MIL-STD-1261, class 3. The workmanship specimens required by MIL-STD-1261 shall be samples simulating all welding conditions.

3.15.4 Bolted connections. Boltholes shall be accurately punched or drilled and shall have the burrs removed. Washers or lock washers shall be provided where necessary. All bolts, nuts, and screws shall be correctly torqued. Matching thread areas securing bolts or capscrews shall be of sufficient strength to withstand the tensile strength of the bolt.

3.15.5 Machine work. Tolerances and gages for metal fits shall conform to the limitations specified on the applicable drawings.

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4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items must meet all section 3 and 5 requirements. The inspections in section 4 are the minimum to be used to demonstrate compliance. Sampling in quality conformance inspection does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.

4.1.2 Component and material inspection. The contractor is responsible for insuring that components and materials used are manufactured, examined, and tested in accordance with referenced specifications, standards and drawings, as applicable.

4.1.3 Parts and components. Parts and components detailed on the drawings shall be inspected in accordance with the quality assurance provisions (QAP) shown on the drawings. The drawings specify the characteristics requiring QAP inspection, the sampling plan, and the basis for acceptance and rejection (see 6.5).

4.2 Classification of inspections. Inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.3).
b. Quality conformance inspection (see 4.4).
c. Inspection of packaging (see 4.6).

4.3 First article inspection.

4.3.1 Examination. The erection equipment, maintenance equipment, or separately procured components as applicable shall be examined for the defects specified in 4.5.1. Presence of one or more defects shall be cause for rejection.

4.3.2 Tests. The erection equipment, maintenance equipment, or separately procured components as applicable shall be tested as specified in 4.5.2. Failure of any test shall be cause for rejection.

4.3.2.1 Hydrualic-system components. Prior to installation of the hydraulic-system components into the first article, the components shall be tested as specified in 4.5.2. Failure of any test shall be cause for rejection.

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4.4 Quality conformance inspection.

4.4.1 Unit of product. For the purpose of inspection the erection equipment shown on TA13217E2171, the maintenance equipment shown on TA13217E2172, or individual components of either of the data lists as specified (see 3.1) shall be considered a unit of product.

4.4.2 Sampling. Sampling for examination and tests of erection equipment, maintenance equipment, and separately procured components shall be in accordance with MIL-STD-105.

4.4.3 Examination.

4.4.3.1 Individual. Each component of the erection equipment, maintenance equipment and each separately procured component shall be examined for defect 101 as specified in 4.5.1. Presence of one or more defects shall be cause for rejection.

4.4.3.2 Samples. Samples selected in accordance with 4.4.2 shall be examined for defects 102 through 202 as specified in 4.5.1. AQL shall be 2.5 percent defective for major defects and 4.0 percent defective for minor defects.

4.4.4 Tests.

4.4.4.1 Individual. Each hydraulic cylinder shall be tested as specified in 4.5.2.1, 4.5.2.2, and 4.5.2.3(d). Failure of any test shall be cause for rejection.

4.4.4.2 Samples. Samples selected in accordance with 4.4.2 shall be tested as specified in 4.5.2.1; 4.5.2.2; 4.5.2.3(a), (b), (c), and (e); 4.5.2.4; and 4.5.2.5. AQL shall be 1.0 percent defective.

4.5 Inspection procedures.

4.5.1 Examination. The erection equipment, maintenance equipment, or separately procured components shall be examined as specified herein for the following defects:

Major

101. Any part or component not in accordance with the QAP requirements as shown on the drawings.
102. Materials not as specified.
103. Any drawing requirement, other than those identified by QAP data, not as specified.
104. Age of hydraulic hose not as specified.
105. Components missing or not as specified.
106. Assembly incomplete or incorrect.
107. Qualification of welders not as specified.

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108. Welding not as specified.
109. Workmanship not as specified.
110. Bolted connections not as specified.
111. Materials are not resistant to corrosion or deterioration or treated to be made resistant to corrosion or deterioration for the applicable storage and operating environment.
112. Dissimilar metals as defined in MIL-STD-889 are not effectively insulated from each other.
113. Contractor does not have documentation available for identification of material, material finishes or treatments.

Minor

201. Cleaning, treatment, and painting not as specified.
202. Identification and other required markings not as specified.

4.5.2 Tests.

4.5.2.1 Test conditions. All tests, except 4.5.2.7, shall be conducted at an ambient temperature between +40° F and 90° F. Hydraulic oil temperature shall not exceed +190° F at the pump discharge during testing. The equipment shall operate as specified herein without maintenance.

4.5.2.2 Cleanliness of hydraulic components. Prior to assembling the hydraulic system, clean and flush each hydraulic component. Prior to flushing, position each cylinder in such a manner that entrapment of contaminants is held to a minimum. Presence of particles on a 100-micron filter screen shall constitute failure of this test.

4.5.2.3 Proof pressure. Subject the following components to a proof pressure of 15,000 psi ±100 psi for 5 minutes:

a. 13217E2206 - 1 and 2 - Hose Assembly.
b. 13217E2196 - 1 and 2 - Hose and Coupling Assembly Female.
c. 13217E2205 - 1 and 2 - Hose and Coupling Assembly Male.
d. 13217E2195 - Cylinder and Assembly Hydraulic Double Acting.
e. Individual components detailed on the drawings, such as, but not limited to, nipples, flow control valves, bushings, quick-disconnect couplings and pressure gages.

The proof pressure shall be applied alternately to each port of the double acting hydraulic cylinder assembly (Drawing 13217E2195).

4.5.2.4 Cable tensioning assembly. Assemble the cable tensioning assembly as specified in 3.4.4 and cycle the assembly a minimum of 25 complete cycles. After completion of 25 cycles, subject the assembly to a pressure of 9,500 psi ±100 psi and hold the pressure for 20 minutes. Nonconformance to 3.4.4 shall constitute failure of this test.

4.5.2.5 Hydraulic hand pump. The hydraulic hand pump shall be tested for conformance with performance criteria shown on the drawings. Failure to conform to the drawing requirements shall constitute failure of this test.

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4.5.2.6 First article performance.

4.5.2.6.1 Mission. Cable reinforcement erection and maintenance equipment components selected as specified in 4.3 shall be tested for a total of 24 missions. A mission shall consist of the following:

a. Load the equipment on the dump truck in accordance with instructions contained in TM 5-277.
b. Transport the equipment in accordance with the following schedule:

(1) Ten miles paved road.
(2) Ten miles secondary road.
(3) Five miles cross country (Aberdeen Proving Ground, Course Perryman No. 3 or equivalent).

c. Off load the erection equipment at the site and install the cable reinforcement equipment on the M-2 Panel Bridge using the erection equipment.
d. During the 88 crossings of the bridge with the Military Load Class (MLC) 60 vehicles use the maintenance equipment to check or adjust cable tension required for the span erected (see TM 5-277). Checks and adjustments, if required, shall be made after 22, 44, 66, 88 crossings of the MLC 60 loads.

4.5.2.6.2 Failure criteria. During performance of the 24 missions the erection and maintenance equipment shall be used in accordance with TM 5-277 and shall perform without evidence of the following defects:

a. External leakage of the hydraulic system components.
b. Evidence of permanent deformation or failure of any component.
c. Material defects, cracks, open joints or weld deterioration.
d. Inability of hydraulic system to provide cable tension required.

4.5.2.7 Hydraulic system environmental test. The hydraulic equipment, as follows, shall be subjected to the tests contained in 4.5.2.4 under temperature conditions of +125° F ±5° F and -25° F ±5° F:

a. Cable tensioning assembly.
b. Maintenance equipment.

4.5.2.7.1 Failure criteria. Failure of equipment to function at the temperature limits specified or evidence of external leakage of a hydraulic component or permanent deformation or damage shall constitute failure of the test.

4.6 Inspection of packaging.

4.6.1 First article pack inspection.

4.6.1.1 Examination. Examine the first article pack for the defects specified in 4.6.2.3. Presence of one or more defects shall be cause for rejection.

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4.6.1.2 Test. The first article pack of crated erection equipment, maintenance equipment, or separately procured components for level A shall be subjected to the railcar impact test in accordance with MIL-STD-1186, Appendix A. At the conclusion of the first article pack test, the pack shall be examined. Any shifting of contents, loosening or breaking of holddowns, ties, stays, blocking or bracing, or any visual damage to the equipment or the chests and boxes shall constitute failure of the first article pack and shall be cause for rejection.

4.6.2 Quality conformance inspection of pack.

4.6.2.1 Unit of product. For the purpose of inspection a completed pack prepared for shipment shall be considered a unit of product.

4.6.2.2 Sampling. Sampling for examination shall be in accordance with MIL-STD-105.

4.6.2.3 Examination. Samples selected in accordance with 4.6.2.2 shall be examined for the following defects. AQL shall be 2.5 percent defective.

114.	Unfinished exterior metal surfaces not coated with preservative as specified for level A.
115.	Machined surfaces not coated with preservative and wrapped and secured as specified for level A.
116.	Cable tensioning assembly not preserved as specified for level A.
117.	Hydraulic maintenance equipment not preserved as specified for level A.
118.	Consolidation not as specified for level A and commercial.
119.	Packing not as specified for level A and commercial.
120.	Blocking, bracing, and anchoring not in accordance with the referenced documents for level A.
121.	Marking missing, illegible, incorrect, or incomplete for level A or commercial.

5. PACKAGING

5.1 First article pack. The contractor shall furnish a first article pack for examination and test within the time frame specified, to prove prior to starting production that the applied preservation, packing, and marking comply with the packaging requirements of this specification. Examination and test shall be as specified in section 4 and shall be subject to surveillance and approval by the Government (see 6.4). The first article pack may be accomplished utilizing either the first article model or production item. If the first article model is utilized, any preservation and packing shall be removed by the contractor at no expense to the Government, when requested by the Government to facilitate comparison between the first article model and the production items.

5.2 Preservation. Preservation shall be level A or commercial as specified (see 6.2).

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5.2.1 Level A.

5.2.1.1 Preservatives. Preservatives specified herein shall conform to the applicable specification listed in and shall be applied in accordance with MIL-P-116.

5.2.1.2 Unprotected surfaces.

5.2.1.2.1 Unfinished surfaces. Unpainted or unfinished exterior metal surfaces of components requiring the application of a contact preservative in accordance with MIL-P-116 shall be coated with type P-1 preservative.

5.2.1.2.2 Machined surfaces. Exposed machined metal surfaces shall be coated with type P-6 or P-11 preservative. The coated surfaces shall be wrapped or covered with barrier material conforming to MIL-B-121, grade A, type I or II, class 2, and material shall be secured in place with tape conforming to PPP-T-60, type IV.

5.2.1.3 Cable tensioning assembly.

5.2.1.3.1 Hydraulic cylinders. Each hydraulic cylinder shall be filled with hydraulic oil (see 3.3.3) and the piston fully retracted into the cylinder and the cylinders placed in a chest or box as specified (see 3.6 and 5.2.1.5).

5.2.1.3.2 Hose and coupling assembly. Each hose and coupling assembly shall be coiled and secured with cotton tape or twine and placed in a chest or box as specified (see 3.6 and 5.2.1.5).

5.2.1.3.3 Hydraulic power unit. Each hydraulic power unit supply reservoir shall be filled with hydraulic oil (see 3.3.3) and placed in the chest or box as specified (see 3.6 and 5.2.1.5).

5.2.1.4 Hydraulic maintenance equipment.

5.2.1.4.1 Manual pump assembly. Each manual pump assembly shall be preserved and placed as specified in 5.2.1.3.1.

5.2.1.4.2 Hose and coupling assembly. Each hose and coupling assembly shall be coiled, secured, and placed as specified in 5.2.1.3.2.

5.2.1.5 Consolidation. When chests are provided (see 3.6) the components shall be placed and secured in the appropriate chest and the chest lid secured. When chests are not provided the components shall be placed and secured in close-fitting boxes conforming to PPP-B-601, overseas type, style optional or PPP-B-621, class 2, style optional. The contents shall be blocked, braced, anchored, or cushioned as required to prevent movement or damage. Box closure shall be in accordance with the appendix to the applicable box specification. Strapping shall not be required. When the individual components are procured as replacements or spares they shall not require consolidation but shall be directly packed as specified in 5.3.1.

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5.2.2 Commercial. The equipment shall be preserved to afford protection against deterioration and damage in accordance with ASTM D 3951. Equipment for which chests are provided shall be placed in the applicable chest and the chest lid closed and secured.

5.3 Packing. Packing shall be level A or commercial as specified (see 6.2).

5.3.1 Level A. The equipment contained in the appropriate chests or boxes shall be packed together in a crate conforming to MIL-C-52950, type V, style A, nondemountable. The chests or boxes shall be secured to the crate base in a manner to prevent movement or damage in accordance with the appendix to the crate specification. When individual components of the erection and maintenance equipment are procured as replacements or spares, the components shall be packed together in a close-fitting box conforming to PPP-B-621, class 2, style optional or PPP-B-601, overseas type, style optional. Box closure and strapping shall be in accordance with the appendix to the applicable box specification. Strapping shall conform to QQ-S-781, class 1, type I or IV, finish B.

5.3.2 Commerical. The equipment preserved as specified in 5.2, shall be packed in a single shipping container in accordance with ASTM D 3951.

5.4 Marking.

5.4.1 Military. Marking for level A shall be in accordance with MIL-STD-129.

5.4.2 Commercial. Commercial marking shall be in accordance with ASTM D 3951. In addition the shipping container shall be marked with weight and cube data.

6. NOTES

6.1 Intended use. The erection and maintenance equipment is intended to be used to attach the cable reinforcement equipment covered by MIL-E-52714(ME) to the triple single configuration of the M-2 panel bridge (Bailey type). The equipment also provides the means of checking and adjusting the post tension in the bridge.

6.2 Ordering data. Procurement documents should specify the following:

a. Title, number, and date of this specification.
b. Whether the contractor shall furnish all the erection equipment on TA13217E2171, all the maintenance equipment on TA132217E2172, or separately procured components selected from either of the two data lists (see 3.1).
c. Time frame required for submission of the first article and number of units of equipment or separately procured components required (see 3.2).
d. When the Goverment will conduct any or all of the first article examination and tests. When the Government will conduct some or all of the first article examination and tests, the contracting officer should specify which examination and tests will be conducted by the Government and which examination and tests shall be conducted by the contractor (see 3.2).

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e. When first article inspection is required and the number of units of equipment or separately procured component, to be furnished (see 3.2).
f. When carrying chests are not required (see 3.6).
g. Time frame for submission of the first article pack (see 5.1).
h. Degree of preservation and degree of packing required (see 5.2 and 5.3).

6.3 First article. When a first article inspection is required, the items should be an initial production item. The first article should consist of one or more units. The contracting officer should include specific instructions in acquisition documents regarding arrangements for examinations, tests and approval of the first article test results and disposition of the document's first articles.

6.4 First article pack. Any changes or deviations of production pack from the approved first article pack will be subject to the approval of the contracting officer. Approval of the first article pack will not relieve the contractor of his obligation to preserve, pack, and mark the equipment and components in accordance with this specification.

6.5 Quality assurance provisions. The contractor shall maintain complete records of all examinations and tests performed to verify the requirements of classified QAP characteristics. The records shall include as a minimum, lot size, sample size, drawing requirement, actual measurement, number and type of deficiencies found, quantity approved, quantity rejected, and corrected action taken when applicable.

6.5.1 Definition. A QAP is a contractual requirement that supplements section 4 of the specification. QAP indicates the minimum requirements which must be inspected on the product drawings to verify the design objectives of the product and assure interchangeability of repair parts.

6.6 Data requirements. The contracting officer should include requirements for such data as technical publications, instructional materials, illustrated parts lists, and contractor's maintenance manual to be furnished with the cable reinforcement equipment.

6.7 Provisioning. The contracting officer should include provisioning requirements for repair parts and maintenance tools as necessary (including special tools), and instructions regarding shipment of the equipment covered by this specification.

6.8 Recycled material. It is encouraged that recycled material be used whenever practical as long at it meets the requirements of the specification.

Custodian:
Army - ME

Preparing activity:
Army - ME

Project 5420-A346

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-E-52715B(ME)	Erection & Maintenance Equipment for Cable Reinforcement Set For Panel Bridge, M-2 (Bailey

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
	☐ USER
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ MANUFACTURER
	☐ OTHER *(Specify)*: _____

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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